Aaron Lane

Bioinformatician | Researcher | Former Educator
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Summary

Bioinformatician and former educator completing a Master's in Biological Data Science with research experience at La Jolla Institute for Immunology. Skilled in statistical analysis, predictive modeling, and data pipeline automation. Published scientific author with strong communication skills. Developing skills in advanced data workflow orchestration.

Skills

Programming: Python, SQL (MySQL), Bash
Data Science & Engineering: Pipeline development, Data cleaning and validation, Predictive modeling, Machine learning, Statistical analysis, Automated data collection (Flask, GCP, BigQuery), Feature engineering
Visualization: Power BI, Matplotlib, Seaborn
Communication: Dashboarding, Scientific writing, Presentations, Cross-functional collaboration, Bilingual – Spanish (C1)

Education

M.S. Biological Data Science – Arizona State University | GPA: 4.0 | Jan 2024 - Dec 2025
M.A. Bilingual Education – University of California, Davis | GPA: 3.9 | Sept 2020 - June 2022
B.S. Biological Sciences – University of California, Santa Barbara | GPA: 3.1 | Sept 2015 - June 2019

Projects

Longitudinal Flu Vaccine Response Dynamics – First Author, *Vaccine* (2025)

- Analyzed data from ~1500 individuals across 12 studies to identify immune-response phenotypes.
- Predicted long-term response trajectories with accuracy similar to assay noise; led analysis, figure design, and publication.

SERENE Mental Health App - ASU Research Project, *in progress*

- Designing a full-stack mobile application with React Native that gathers survey, image, and GPS data to study mental health–nature relationships, with cloud-based computer vision and BigQuery pipelines for automated analysis.
- Leading analysis to identify areas that need enhanced blue and green infrastructure based on mental health.

Flu Search Trends Predictor Dashboard – Independent Project

- Integrated CDC FluView and Google Trends data to predict regional influenza activity across 2021-2025 seasons.
- Built an automated Python → BigQuery → Power BI pipeline with Cloud Run scheduling to feed a live dashboard.

Heat Behind Bars – ASU Master's Capstone

- Quantified heat-related mortality risk in Arizona prisons using climate and facility infrastructure data.
- Co-authored manuscript in collaboration with ASU researchers.

Notebook Comment Translator – Open Source Tool

- Developed a Python utility that translates Jupyter markdown and comments between English ↔ Spanish.

Professional Experience

Bioinformatics Specialist I – La Jolla Institute for Immunology | May 2024 – Aug 2025

- Analyzed influenza serology data to model the vaccine response; published and presented findings.
- Developed standardized analysis pipelines and collaborated across global research teams to curate large datasets.

Chemistry Teacher – Vacaville High School | Aug 2022 – Jun 2023

- Collaborated with colleagues to revamp chemistry program to include lab practicals and inquiry-based learning.
- Taught chemistry and data analysis concepts; created hands-on labs and assessments.
- Managed chemical storeroom, including waste storage, organization, and orders.

ESL Science Teacher – Oakland Unified School District | Aug 2021 – May 2022

- Adapted biology instruction for English learners; integrated bilingual teaching methods.
- Coordinated with EMTs to certify ~30 students in CPR and first aid training.

Science Teacher - Eureka School of Santa Barbara | Aug 2019 - Dec 2019

- Designed and delivered curricula for four small multi-level classes (2-6 students)
- Differentiated instruction to meet varied learning needs and communicated with parents to ensure student success.